

Production-Readiness Audit + Remediation Plan — 2026-08-09

Status: LIVE DOCUMENT — being updated as background verification (constitutional sweep, containerized device gate) completes. Read docs/CONTINUATION.md §0 for the current pointer; this file is the point-by-point punch list behind it.

Why this exists. The operator asked for a full anti-bluff audit against the constitution, an inventory of every open workable item, identification of every gap/inconsistency/danger zone, and a systematic plan to close each one with real, executed, non-bluff evidence. This document is that plan. Every claim below is either (a) independently verified against the actual repo/DB state in this session (not trusted from a subagent's summary alone), or (b) explicitly marked PENDING with what will confirm it.

Session interruption note (forensic, §6.M). The Claude Code process was killed mid-session twice (once ~15:55, once ~18:5x) while long-running verification agents were in flight. Both times uptime confirmed continuous host uptime — these were process-level interruptions, not host power events. No git-tracked work was lost; one in-flight device-gate attempt had to be redone from a very early stage (it had only logged its runner selection before being killed). Lesson applied: long-running verification (the constitutional sweep, the device-gate Challenge run) now executes as direct background shell commands from the main session rather than nested inside subagents, which is more resilient to this class of interruption.

1. Executive summary

Category	Count	Detail
Confirmed-fixed this session	3	credential leak (§6.H/§6.AC), markdown-export-sync gate, coverage-ledger gate
	8	

Category	Count	Detail
Confirmed already-fixed-in-source, pending device verification		LVA-083, LVA-084, LVA-085, LVA-086, LVA-087, LVA-088, LVA-089, LVA-090
Genuinely unconfirmed, needs device/operator input	1	LVA-091 (app-ID co-mingling)
Doc-staleness corrections applied	4	submodule pin table, LVA-6 status, LVA-7 status, Firebase-token-resolution claim
Structurally operator-gated (no agent can close these)	4	Firebase CI token rotation, 2× GitLab mirror push, LVA-008 upstream Android bug
Verification runs in flight	2	full 54-gate constitutional sweep; 6-Challenge containerized device gate

Bottom line so far: nothing found this session indicates the product is fundamentally broken — the P0 "search is completely unusable" reports (LVA-083/084) already have real root-caused fixes in source from a 2026-06-25/06-26 cycle; they were simply never verified on a device before being left as open tickets. The constitutional-gate debt is almost entirely mechanical staleness (regenerate a ledger, regenerate doc exports), not missing engineering. The remaining real risk concentrations are: (1) whether the device-verifiable fixes actually pass on a real emulator (verification in flight), (2) two credential/mirror items that require the operator's hands, and (3) one confirmed upstream Android framework bug (LVA-008) that Lava's own code cannot fix.

2. Fixed and independently verified this session

2.1 Credential leak — `OnboardingViewModel.kt:659` (§6.H / §6.AC)

`LoginResult` is a Kotlin data class with a `raw sessionToken: String?` field and no custom `toString()`; `logger.d { "...result=$loginResult" }` therefore wrote the session token to `logcat` on every login. **Verified by reading the class definition**

directly (not trusted from any agent claim). Fixed: log now emits only state/hasSessionToken/hasCaptcha. Not yet committed (bundled into the consolidated commit — see §6).

2.2 markdown-export-sync gate

Was FAILING (8 stale HTML/PDF exports: CHANGELOG.md, docs/CONTINUATION.md, docs/Fixed.md, docs/Issues.md). Regenerated; **re-ran the check script directly myself**: checked 265 in-scope .md file(s); 0 problem(s). Confirmed PASS.

2.3 coverage-ledger gate (§11.4.25)

Was stale (last generated 2026-07-26, months of module changes since). Regenerated via scripts/generate-coverage-ledger.sh (66 rows: 49 covered / 9 partial / 8 gap) and **re-ran scripts/check-coverage-ledger.sh --strict directly myself**: exit 0, "ledger is well-formed, complete, and fresh". The 8 remaining gap rows (core/analytics-firebase, core/apiengine, core/applink, and 5 HelixQA-sibling submodules: doc_processor/llm_orchestrator/llm_provider/llms_verifier/vision_engine) are a real, separately-tracked item — see §4.3 (LVA-019).

2.4 Documentation staleness in docs/CONTINUATION.md

- §3 submodule pin table: 22 of 24 rows were stale (some by 2+ months); regenerated from a live git submodule status + per-submodule fetch sweep.
 - Flagged (not silently fixed): 2 real GitHub↔GitLab mirror mismatches (panoptic, tracker_sdk — GitLab is a real ancestor of GitHub); fixing requires a git push to a shared remote, which needs operator sign-off (see §5.2).
 - LVA-6 and LVA-7: doc said "OWED"/"IN PROGRESS"; DB ground truth (docs/workable_items.db) says both Completed (→ Fixed.md). Corrected.
 - Firebase CI token: doc claimed "RESOLVED 2026-05-31"; DB (LVA-5) says Operator-blocked as of 2026-06-30, and the incident file the doc cites doesn't exist in the tree. Marked honestly UNCONFIRMED rather than picking a side — see §5.1.
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3. Confirmed already-fixed-in-source, pending only device verification

All 8 below trace to the same operator QA-video cluster (.lava-ci-evidence/video-analysis/2026-06-25-lava-issues-video.md). For each, I (or a dispatched agent whose specific file:line claims I spot-verified — see §3.9) read the actual production code and the actual reproduce-first Challenge Test before accepting the classification. None required a new code fix; all trace to a real prior fix commit.

Ticket	Symptom	Fix location	Covering Challenge
LVA-083 (P0)	Search returns zero results / "Something went wrong"	SearchInputViewModel.kt:104-109 — explicit onboarded-provider set, never the old null-means-all sentinel	Challenge58SearchReturn
LVA-084 (P0)	Onboarded provider (YTS) not the one Search uses	same fix as LVA-083 (observeAll().filter { searchEnabled && isEnabled })	Challenge59SearchUsesC
LVA-085 (P1)	Provider chip labels raw/lowercased	ProviderCatalogRepository.kt:217 friendlyDisplayName() humanizes blank/id-equal display names at the adapter boundary	Challenge61ResultsChip
LVA-086 (P1)	No empty/loading state, looked hung	SearchResultScreen.kt:347-358 renders a loading spinner while streaming with zero items; handleStreamEnd() always terminates into Content/Empty/Error	Challenge62SearchEmpty
LVA-087 (P2)	Welcome says "4 providers", picker shows ~12	OnboardingViewModel.kt:472 sets welcomeProviderCount = null when apiSelectionEnabled (true in production), so the stale exact count never renders	Challenge63WelcomeCour
LVA-088 (P2)	Cloud API preset mislabeled	ApiSelectionStep.kt:254,376 gives cloud-default rows an explicit "Cloud / remote server" subtitle	(covered by onboarding)

Ticket	Symptom	Fix location	Covering Challenge
LVA-089 (P2)	"On this network" mDNS API shown as raw IP, no name	ApiSelectionStep.kt:162,329,385-388 resolves the mDNS instance name from a host:port-keyed map when present	Challenge65MdnsShowsFr
LVA-090 (P3)	"Select all" silently enabled auth- requiring providers	OnboardingViewModel.kt:592-622 onToggleAllProviders() only auto-selects requiresNoCredentials() providers	Challenge66SelectAllDo

3.9 Independent verification note

I personally read `SearchInputViewModel.kt` in full (§3 LVA-083/084) and `Challenge58SearchReturnsResultsTest.kt` in full (confirmed it is a genuine, non-bluff test: real `MainActivity`→`ViewModel`→`SDK`→`OkHttp` path, `MockWebServer` only at the socket boundary, primary assertion on a rendered result row, documented reproduce-first mutation protocol). I spot-checked LVA-090's `onToggleAllProviders` directly against the reporting agent's claim and found it accurate, including the exact commit (a300b3b0). The remaining LVA-085/086/087/088/089 citations are trusted from the dispatched agent's report on the strength of that spot-check plus the specificity of every citation (exact file:line, exact function names) — a standard I judge acceptable but weaker than full independent re-reading of all 6.

3.10 One unresolved side-observation (not a ticket, logged here so it isn't lost)

`ProvidersStep.kt:74`'s "Select all" checkbox label computes `allSelected` from *all* providers, while the `ViewModel`'s toggle-direction decision only considers no-credential-reachable providers. When credential-required providers exist and stay unselected, the checkbox may never flip to "Deselect all" even once every reachable provider is selected. Does NOT reproduce LVA-090's reported symptom (auth providers being silently enabled) — flagged as a possible follow-up, not fixed this session, not blocking.

4. Genuinely open, needs more than what's in this repo

4.1 LVA-091 (P3) — App-ID co-mingling, UNCONFIRMED

Source inspection across 3 independent layers (`applicationId/`
`applicationIdSuffix` in `app/build.gradle.kts`, per-build-type `app_name`
string override, per-build-type launcher icon override) shows strong
evidence debug and release ARE already distinguishable (different
package id, different label, different icon color). The ticket's own title
says UNCONFIRMED. **Closure requires an on-device check**
(`adb shell pm list packages | grep lava` + a visual launcher check) —
folded into the device-gate run in §5.3 if time permits; otherwise this is
the one item where "needs a real device, not just source reading" is a
legitimate, non-bluff blocker.

4.2 LVA-008 (P1) — nav-teardown crash — confirmed upstream defect, not Lava's

8 client-side candidate fixes have been device-falsified over prior
cycles; a minimal repro is authored for upstream `androidx-navigation`.
This is not something any amount of further Lava-side work will close
— it needs either an `androidx-navigation` fix to land upstream, or an
operator decision to ship with the existing `LenientTeardownRule` test-
harness workaround (which only suppresses the teardown-time
exception in tests, not in the shipped app) permanently. **Operator
decision owed**, not additional engineering.

4.3 LVA-019 (P1) — coverage-ledger gap rows

8 modules show gap status in the freshly-regenerated ledger: `core/`
`analytics-firebase`, `core/apiengine`, `core/applink`, plus 5 HelixQA-
sibling Go submodules (`doc_processor`, `llm_orchestrator`, `llm_provider`,
`llms_verifier`, `vision_engine`). The gate itself passes (gaps are tracked,
not blocking), but per the "no work forgotten" mandate these are real
coverage holes worth a follow-up session — not addressed here
because closing 8 modules' test coverage properly is multi-session
scope, not a mechanical fix.

4.4 LVA-013 (P0 per body) — missing §6.Z device evidence for already-shipped 1080 / api-app 24

Folded into the device-gate work in §5.3 as a secondary objective once the 1081 Challenge set completes, resources permitting.

5. Structurally operator-gated (cannot be closed by an agent, full stop)

5.1 Firebase CI token — contradictory status, needs operator confirmation

One commit claims rotation happened 2026-05-31; the workable-items DB says LVA-5 is still Operator-blocked as of 2026-06-30; .env was touched 2026-07-03 (after both dates) so a rotation may have happened without the tracker being updated. **I cannot determine which is true from file evidence, and only the operator can actually rotate LAVA_FIREBASE_TOKEN** (the ticket's own unblock instructions say so explicitly). Action needed: confirm current token validity, or rotate it (`firebase logout && firebase login:ci`), then close LVA-5 in the DB.

5.2 Two GitHub↔GitLab mirror mismatches

panoptic and submodules/tracker_sdk both have a GitLab tip that is a real, unreconciled ancestor of the GitHub tip. Fixing this is a `git push gitlab` per submodule — a shared-remote action I won't take without explicit sign-off per this session's operating constraints. panoptic is additionally missing `install_upstreams.sh` (§11.4.36 gap) — I can add that script without pushing anything if that's useful groundwork; say so if wanted.

5.3 Real-device verification of the 8 already-fixed tickets + the crash regression

This is IN PROGRESS as a background containerized-KVM Challenge run (see §6) — not operator-gated, just genuinely takes real wall-clock time (cold emulator boot ×6 test classes). Will update this section with PASS/FAIL evidence per test the moment it completes.

6. In-flight verification (background, main-thread, resilient to process kills)

1. **Full 54-gate constitutional sweep** (`scripts/verify-all-constitution-rules.sh`) — re-running from a clean state now that both known failures (§2.2, §2.3) are fixed. Expect a materially better pass count than the last completed run (53/54, single ledger failure, recorded 2026-07-26).
 2. **Containerized device gate**, 6 Challenge Tests against the current build on a real containerized-KVM emulator (this Linux host, confirmed `/dev/kvm` + `podman` available — the exact gate path prior macOS-hosted sessions in this project's history could never reach):
 - Challenge00CrashSurvivalTest (cold-start canary) →
 - Challenge71CategorySelectionDialogBoundedHeightTest (closes the 1.3.15-1081 Crashlytics regression) →
 - Challenge58SearchReturnsResultsTest (closes LVA-083) →
 - Challenge59SearchUsesOnboardedProvidersTest (closes LVA-084) →
 - Challenge60InputResultsChipsAgreeTest →
 - Challenge62SearchEmptyStateTest (closes LVA-086). Evidence lands incrementally at `.lava-ci-evidence/lva014-c00-c71-c58-c59-device-gate/` as each test finishes, specifically so a third process interruption (if it happens) doesn't lose already-completed results.
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7. What "done" looks like for this cycle

Once §6's two runs complete:

1. Update this document + `docs/CONTINUATION.md` with the real PASS/FAIL evidence (not before — no claiming green ahead of the actual run per §6.J/§6.Z).
2. If all 6 Challenges pass: LVA-083, LVA-084, LVA-086 close for real (device-verified, not just source-verified); LVA-085/087/088/089/090 stay source-verified-only unless their own Challenges also got a slot; the 1.3.15-1081 Crashlytics regression closes.
3. If anything fails: that's a genuine finding, not a setback to hide — root-cause it via the same systematic-debugging process used throughout this session before touching the fix.
4. Bundle every working-tree change from this session into one well-documented, Bluff-Audit-stamped commit (credential-fix, CONTINUATION corrections, coverage-ledger, doc exports) — not before the device-gate evidence exists, since the commit should carry real evidence paths, not promises.

5. Present the operator with the short, genuinely-irreducible punch list from §5 — token rotation, 2 mirror pushes, LVA-008 upstream decision, LVA-091 device check — as the honest remainder between "everything an agent can verify is verified" and "100% shipped."